This assignment will not be graded and is only for practice.

Note : This activity is only for practice purpose and it will not be counted towards the Final score.

1) Let $f(x,y)$ be defined in a neighbourhood of (a,b) and $D(x,y) = f_{xx}f_{yy} - f_{xy}^2$. Suppose (a,b) is a critical point of f. Which of the following options is(are) true? ***1 point***

- [ ] If $D(a,b) > 0$ and $f_{xx}(a,b) > 0$, then (a,b) is a point of local minimum.
- [ ] If $D(a,b) < 0$, then (a,b) is a point of local maximum.
- [ ] If $D(a,b) = 0$, then (a,b) may be a point of minimum or maximum or neither.
- [ ] If $D(a,b) < 0$, then (a,b) is a saddle point.

No, the answer is incorrect.

Score: 0

Accepted Answers:

If $D(a,b) > 0$ and $f_{xx}(a,b) > 0$, then (a,b) is a point of local minimum.
If $D(a,b) = 0$, then (a,b) may be a point of minimum or maximum or neither.
If $D(a,b) < 0$, then (a,b) is a saddle point.

2) Choose the correct statements about $f(x,y) = x^4 + y^4$, where $D(x,y) = f_{xx}f_{yy} - f_{xy}^2$. ***1 point***

- [ ] $(0,0)$ is the only critical point.
- [ ] $D(0,0) > 0$ and $f_{xx}(0,0) < 0$ and so $(0,0)$ is a point of local maximum.
- [ ] $D(0,0) = 0$.
- [ ] $(0,0)$ is a point of local minimum.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$(0,0)$ is the only critical point.
$D(0,0) = 0$.
$(0,0)$ is a point of local minimum.

3) Let $f(x,y) = (9x^2 - 1)(1 + 4y)$ be defined on the box $-2 \leq x \leq 3$ and $-1 \leq y \leq 4$. Then which of the following options is (are) true? ***1 point***

- ○ $(0, 1)$ is a point of local minimum.
- ○ $(3, 4)$ is a point of absolute maximum.
- ○ -105 is the absolute minimum.
- ○ $(-2, -1)$ is a point of absolute minimum.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$(3, 4)$ is a point of absolute maximum.

4) Suppose $z = f(x) + yg(x)$ for some polynomial functions f and g. Which of the following options is true? ***1 point***

- ○ $z_{yy} = 0$
- ○ z_{yy} is a non-constant function of x.
- ○ z_{yy} is a non-constant function of y.
- ○ $z_{yy} = 1$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$z_{yy} = 0$

5) Define a function ***1 point***

$$f(x,y) = \begin{cases} \frac{xy(x^2-y^2)}{x^2+y^2} & \text{if } (x,y) = (0,0) \\ 0 & \text{if } (x,y) = (0,0) \end{cases}$$

Choose the set of correct options.

- ☐ $f_{xy}(0,0) = -1$
- ☐ $f_{yx}(0,0) = 1$
- ☐ $f_{xy}(0,0) = f_{yx}(0,0)$
- ☐ $f_x(0,0) = f_y(0,0)$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$f_{xy}(0,0) = -1$
$f_{yx}(0,0) = 1$
$f_x(0,0) = f_y(0,0)$

6) Consider a function $f(x,y,z) = x^3y + yz$. If A is the the Hessian matrix of $f(x,y,z)$ at (1,1,0), ***1 point*** then which of the following options is(are) true?

- ☐ $det(A) = 0$
- ☐ $Rank(A) = 2$
- ☐ $Nullity(A) = 1$
- ☐ $det(A) = -6$
- ☐ $Rank(A) = 3$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$det(A) = -6$
$Rank(A) = 3$

7) Let $f : \mathrm{R}^6 \rightarrow \mathrm{R}$ be a polynomial function. Which of the following options is true? ***1 point***

- ○ Hessian matrix of the function is a symmetric matrix of order 6.
- ○ All possible mixed partial derivatives of second order have the same value at a point of R^6
- ○ Hessian matrix of the function is a symmetric matrix of order 5.
- ○ Hessian matrix of the function is a symmetric matrix of order 4.

No, the answer is incorrect.

Score: 0

Accepted Answers:

Hessian matrix of the function is a symmetric matrix of order 6.

Let S be a rectangular box. Let x, y and z be the length of the sides of the box and volume of the box $V = xyz$. If d is the length of the diagonal of the rectangular box, then $d = \sqrt{x^2 + y^2 + z^2}$. Use this information to answer the questions 8 and 9.

8) If the diagonal is of length 2 units and the volume of the box is the maximum possible, then which of the following options is (are) true? ***1 point***

- ☐ S is a cube.
- ☐ All sides are of different length.
- ☐ $x = \frac{2}{\sqrt{3}}$
- ☐ $y = \sqrt{\frac{1}{3}}$

No, the answer is incorrect.

Score: 0

Accepted Answers:

S is a cube.
$x = \frac{2}{\sqrt{3}}$

9) If the length of the diagonal of the box is 3 units, then which of the followings is the maximum possible volume of the box? ***1 point***

- ○ $\sqrt{3}$ cube units.
- ○ $3\sqrt{3}$ cube units.
- ○ 1 cube units.
- ○ 3 cube units

No, the answer is incorrect.

Score: 0

Accepted Answers:

$3\sqrt{3}$ cube units.

10) Consider a function $f(x,y)$.Let v_1, v_2, v_3 be the critical points of the function f and H_k be the Hessian matrix of the function at the point v_k.Then which of the following options is(are) true? ***1 point***

- ☐ If $H_1 = \begin{bmatrix} 1 & 0 \\ -3 & 2 \end{bmatrix}$, then v_1 is a saddle point.
- ☐ If $H_2 = \begin{bmatrix} 1 & 1 \\ 3 & 2 \end{bmatrix}$, then v_2 is a saddle point.
- ☐ If $H_3 = \begin{bmatrix} 3 & 1 \\ -3 & 1 \end{bmatrix}$, then v_3 is a point of local extremum.
- ☐ If $H_1 = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$, then v_1 is a point of local minimum.

No, the answer is incorrect.

Score: 0

Accepted Answers:

If $H_2 = \begin{bmatrix} 1 & 1 \\ 3 & 2 \end{bmatrix}$, then v_2 is a saddle point.

If $H_3 = \begin{bmatrix} 3 & 1 \\ -3 & 1 \end{bmatrix}$, then v_3 is a point of local extremum.

If $H_1 = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$, then v_1 is a point of local minimum.

11) Consider the following function **1 point**

$f(x, y, z) = x^2y^2 + xz^2$

Which of the following options is (are) true?

- ☐ $(0, y, 0)$, where $y \in \mathrm{R}$ are critical points.
- ☐ $(0, 0, z)$, where $z \in \mathrm{R}$ are critical points.
- ☐ $(1, y, 0)$ where $y \in \mathrm{R}$ are critical points.
- ☐ $(0, y, 1)$, where $y \in \mathrm{R}$ are critical points.
- ☐ $(x, 0, 0)$, where $x \in \mathrm{R}$ are critical points.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$(0, y, 0)$, where $y \in \mathrm{R}$ are critical points.
$(x, 0, 0)$, where $x \in \mathrm{R}$ are critical points.

12) Consider a function $f(x, y, z)$. Let v_1, v_2, v_3 be the critical points of the function f and H_k be the Hessian matrix of the function at the point v_k.Then which of the following options is(are) true? **1 point**

- ☐ If $H_1 = \begin{bmatrix} -1 & 0 & 0 \\ 1 & 1 & -1 \\ 1 & 1 & 1 \end{bmatrix}$, then v_1 is a saddle point.
- ☐ If $H_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$, then v_2 is a saddle point.
- ☐ If $H_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$, then v_3 is a point of local maximum.

- [ ] If $H_1 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$, then v_1 is a point of local minimum.
- [ ] If $H_2 = \begin{bmatrix} -1 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -3 \end{bmatrix}$, then v_2 is a point of local maximum.
- [ ] If $H_3 = \begin{bmatrix} -2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$, then v_3 is a point of local minimum.

No, the answer is incorrect.

Score: 0

Accepted Answers:

If $H_1 = \begin{bmatrix} -1 & 0 & 0 \\ 1 & 1 & -1 \\ 1 & 1 & 1 \end{bmatrix}$, then v_1 is a saddle point.

If $H_1 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$, then v_1 is a point of local minimum.

If $H_2 = \begin{bmatrix} -1 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -3 \end{bmatrix}$, then v_2 is a point of local maximum.

Your score is: 0/12